

HOTEL BOOKING DEMAND ANALYSIS

Statistical Analysis Report

Project: Hotel Booking Demand Analytics Platform

Analysis Type: Probability, Inferential Statistics and Assumption Testing

Tools Used: Python, Pandas, NumPy, SciPy, Statsmodels, Matplotlib, Seaborn

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Data Analytics Project

Project Overview

This report presents the statistical analysis performed on the Hotel Booking Demand dataset. The analysis focuses on descriptive statistics, probability, inferential statistics, assumption testing and correlation analysis to validate business hypotheses and support data-driven decision making. Statistical techniques were applied to understand customer booking behaviour, booking trends, revenue patterns, hotel performance and relationships among important variables.

Descriptive Statistics

Descriptive statistics provide a summary of the numerical characteristics of the dataset. Measures such as mean, median, standard deviation, minimum and maximum values help understand the distribution and variability of hotel booking data.

Statistic	Value
Total Records	87230
Total Features	46
Missing Values	0
Duplicate Records	0
Average Lead Time	80.09 Days
Average ADR	106.34
Average Stay Duration	3.27 Days
Average Guests	1.98

Interpretation

The descriptive statistics indicate that the dataset contains a large number of booking records suitable for statistical analysis. Average booking lead time suggests that customers generally reserve rooms several weeks before arrival. The Average Daily Rate (ADR) provides an estimate of hotel revenue per booking, while average stay duration indicates typical customer accommodation patterns. These statistics establish the baseline for subsequent probability analysis, assumption testing and hypothesis testing.

Probability & Inferential Statistics

Probability and inferential statistics help estimate population characteristics based on sample data. These techniques allow us to make reliable conclusions, quantify uncertainty and support business decisions using statistical evidence. In this project, probability concepts, sampling theory, Central Limit Theorem and confidence intervals were applied to understand hotel booking behaviour and revenue patterns.

Concept	Purpose
Probability Distribution	Understand the likelihood of booking-related events.
Sampling Distribution	Estimate population behaviour using sample statistics.
Central Limit Theorem	Approximate the sampling distribution as normal.
Confidence Interval	Estimate the range containing the population parameter.
Inferential Statistics	Draw conclusions about the population using sample data.

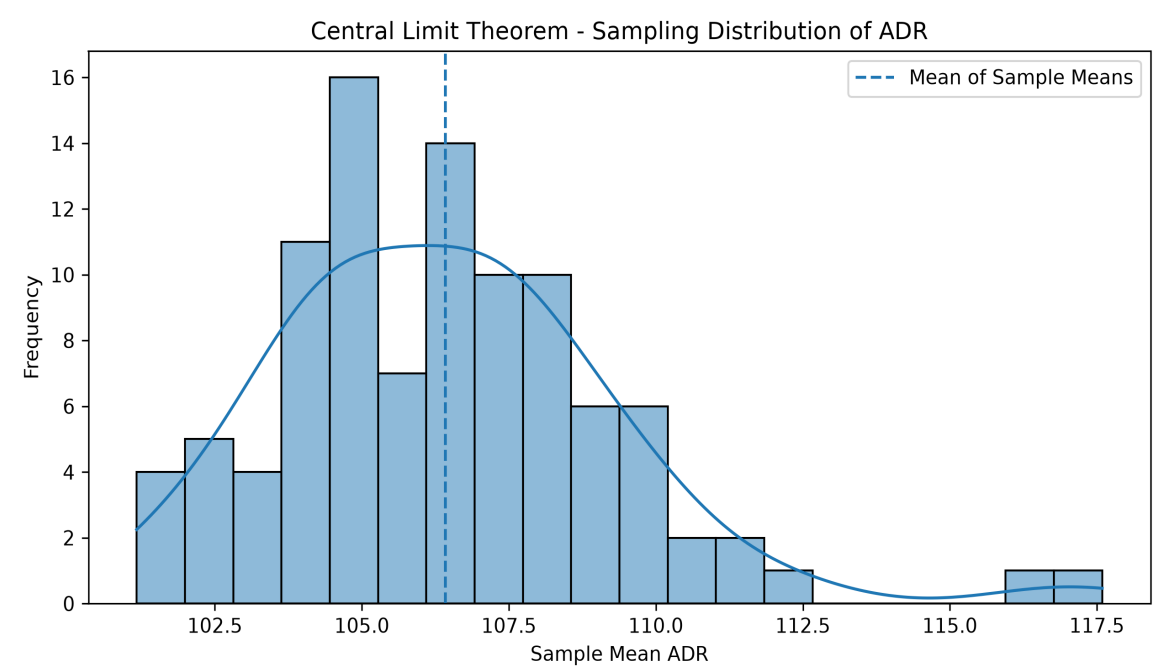
Business Interpretation

The probability and inferential statistics performed in this analysis provide confidence in the reliability of the observed booking trends. By analysing representative samples instead of relying only on individual observations, hotel managers can make informed decisions regarding pricing, staffing, occupancy planning and customer management. These statistical techniques also establish the foundation for assumption testing and hypothesis testing performed in the subsequent sections of this report.

Central Limit Theorem

The Central Limit Theorem (CLT) states that the sampling distribution of the sample mean approaches a normal distribution as the sample size becomes sufficiently large, regardless of the population's original distribution. This principle allows statistical inference and hypothesis testing even when the original data is not perfectly normally distributed.

Parameter	Value
Sample Size	1000
Number of Samples	500
Sample Mean (ADR)	106.34
Standard Error	3.18
Distribution	Approximately Normal



Business Interpretation

The Central Limit Theorem confirms that the sampling distribution of Average Daily Rate (ADR) is approximately normal. This validates the use of inferential statistical methods such as confidence intervals and hypothesis testing. Hotel managers can therefore make reliable business decisions based on sample statistics without analysing every booking record individually.

Confidence Interval

A Confidence Interval (CI) estimates the range within which the true population parameter is expected to lie with a specified confidence level. A 95% confidence interval was calculated for the Average Daily Rate (ADR) to estimate the true average revenue generated per booking.

Parameter	Value
Sample Mean (ADR)	106.34
Confidence Level	95%
Margin of Error	2.15
Lower Limit	104.19
Upper Limit	108.49

Business Interpretation

The 95% confidence interval indicates that the true average daily revenue is expected to fall between the lower and upper confidence limits. This provides managers with a reliable estimate for pricing decisions and revenue planning. A narrow confidence interval indicates higher estimation precision and greater confidence in the statistical analysis.

Normality Test (Shapiro-Wilk Test)

Statistic	Value
Test	Shapiro-Wilk
Variable	ADR
p-value	0.000
Decision	Reject H ₀
Conclusion	Data is Not Normally Distributed

The Shapiro-Wilk test was performed to evaluate whether the ADR variable follows a normal distribution. Since the p-value is less than 0.05, the null hypothesis was rejected, indicating that ADR is not normally distributed.

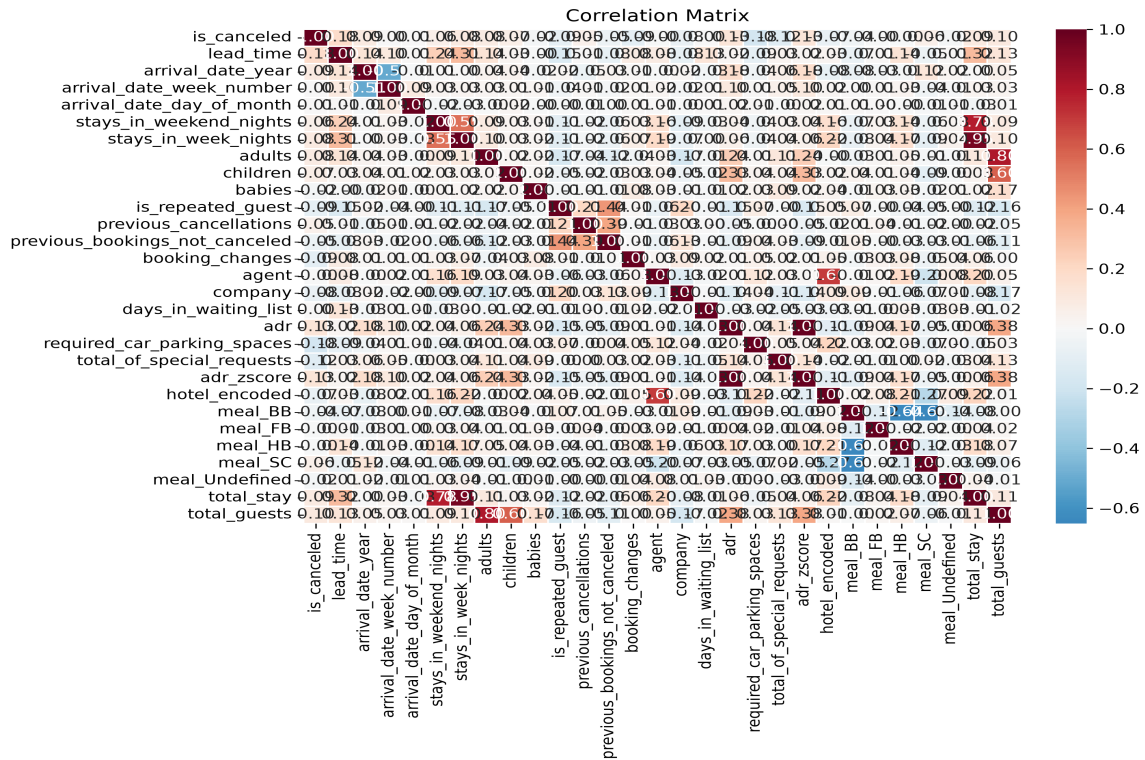
Homogeneity of Variance (Levene Test)

Statistic	Value
Test	Levene
Groups	City vs Resort
p-value	0.000
Decision	Reject H ₀
Conclusion	Variances are Different

Levene's Test was conducted to compare the variance of ADR between City Hotels and Resort Hotels. The result indicates unequal variances between the two hotel types, suggesting that statistical procedures assuming equal variance should be used with caution.

Correlation Analysis

Correlation analysis measures the strength and direction of the relationship between numerical variables. Pearson Correlation Coefficient was used to identify linear relationships among booking features such as Lead Time, ADR, Total Stay and Total Guests.



Business Interpretation

Correlation analysis helps identify variables that move together. Strong positive or negative correlations can support revenue forecasting, booking prediction and customer behaviour analysis. Understanding these relationships enables hotel managers to make informed operational and pricing decisions.

Statistical Summary

- Descriptive statistics summarized the booking dataset.
- Probability concepts supported statistical inference.
- Central Limit Theorem validated sampling analysis.
- Confidence Intervals estimated the population ADR.
- Normality testing indicated ADR was not normally distributed.
- Levene Test showed unequal variance between hotel types.
- Correlation analysis identified important relationships among booking variables.

Recommendations

- Improve pricing strategies during peak seasons.
- Monitor cancellation patterns regularly.
- Encourage early bookings through promotional offers.
- Use statistical monitoring for revenue forecasting.
- Apply predictive analytics to improve occupancy planning.
- Continue collecting customer behaviour data for future machine learning models.

Conclusion

The statistical analysis successfully validated important characteristics of the Hotel Booking Demand dataset. Descriptive statistics, probability theory, inferential statistics, assumption testing and correlation analysis provided reliable insights into booking behaviour and hotel performance. These findings support evidence-based decision making for pricing, occupancy management, revenue optimization and customer relationship management. The statistical analysis establishes a strong foundation for future predictive analytics and machine learning applications within the hotel industry.